Forecasting with stationary models II

This version: October 29, 2026

ECMA 31150, Spring 2026

University of Chicago

Department of Economics

Mikkel Plagborg-Moller

Plan for this slide deck

1. Autoregressive (AR) models
2. Estimating/testing AR parameters
3. Selecting the number of lags

AR(1) model

- Let's return to the autoregressive model we saw in the last slide deck:

$$y_t = c + \rho y_{t-1} + u_t, \tag{$\dagger$}$$

  where c and ρ are constants.

- Assume u_t is white noise satisfying

$$E[u_t \mid y_{t-1}, y_{t-2}, \dots] = 0.$$

  We call u_t the "shock" or **innovation** – the unforecastable part of y_t.

  - In many cases, we'll assume the stronger condition that u_t is i.i.d. with mean zero.

- This model is an autoregression of **order** 1 (one lag): AR(1).

- When can we find a covariance stationary solution to the stochastic difference equation ($\dagger$)?

AR(1) model: stationarity

- Since u_t and y_{t-1} are uncorrelated (why?),

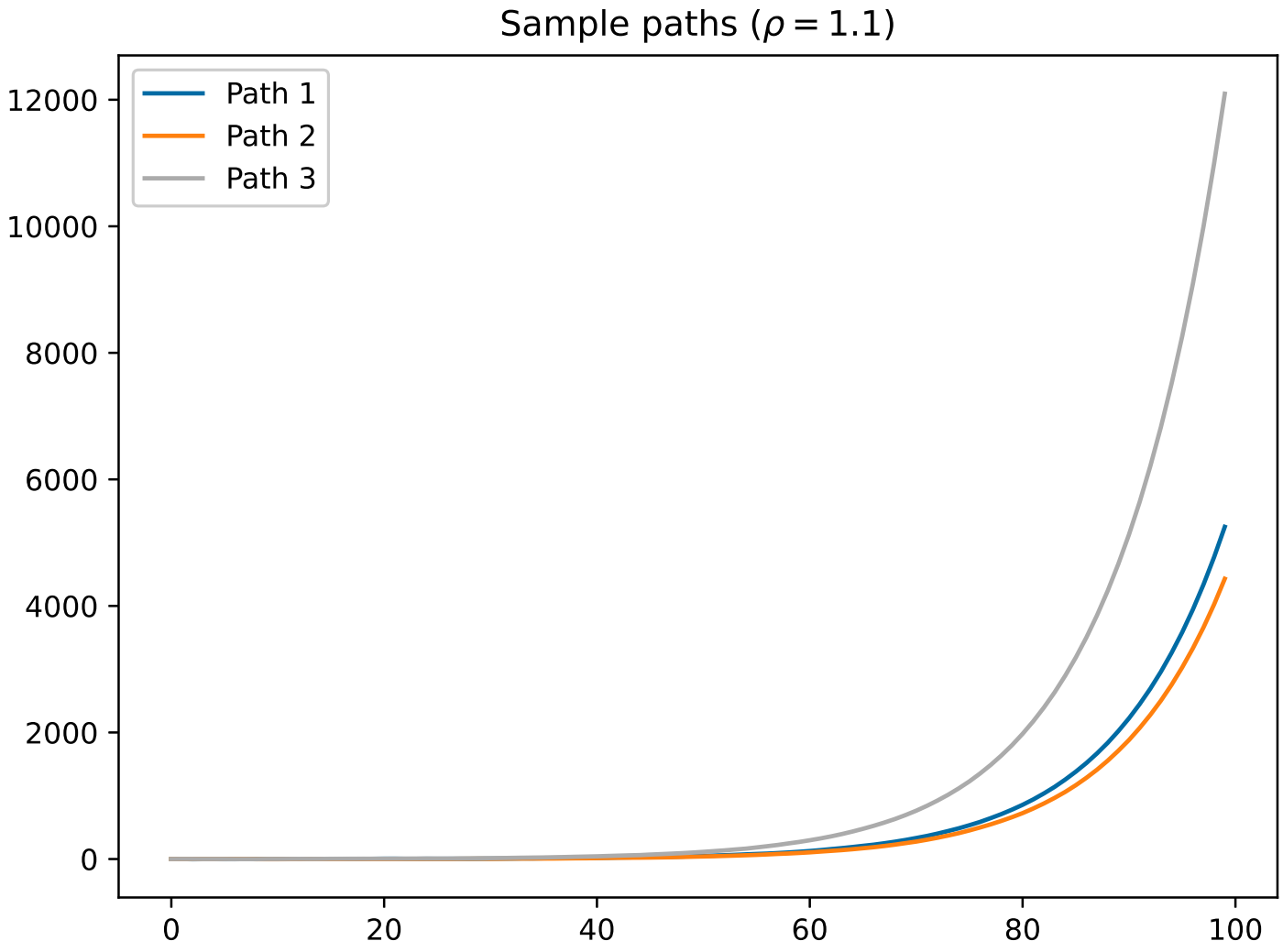
$$\text{Var}(y_t) = \text{Var}(c + \rho y_{t-1} + u_t) = \rho^2 \,\text{Var}(y_{t-1}) + \text{Var}(u_t).$$

- If $\{y_t\}$ is covariance stationary, then

$$\text{Var}(y_t) = \rho^2 \,\text{Var}(y_t) + \text{Var}(u_t) \quad \Longrightarrow \quad \boxed{\text{Var}(y_t) = \frac{\sigma_u^2}{1-\rho^2}},$$

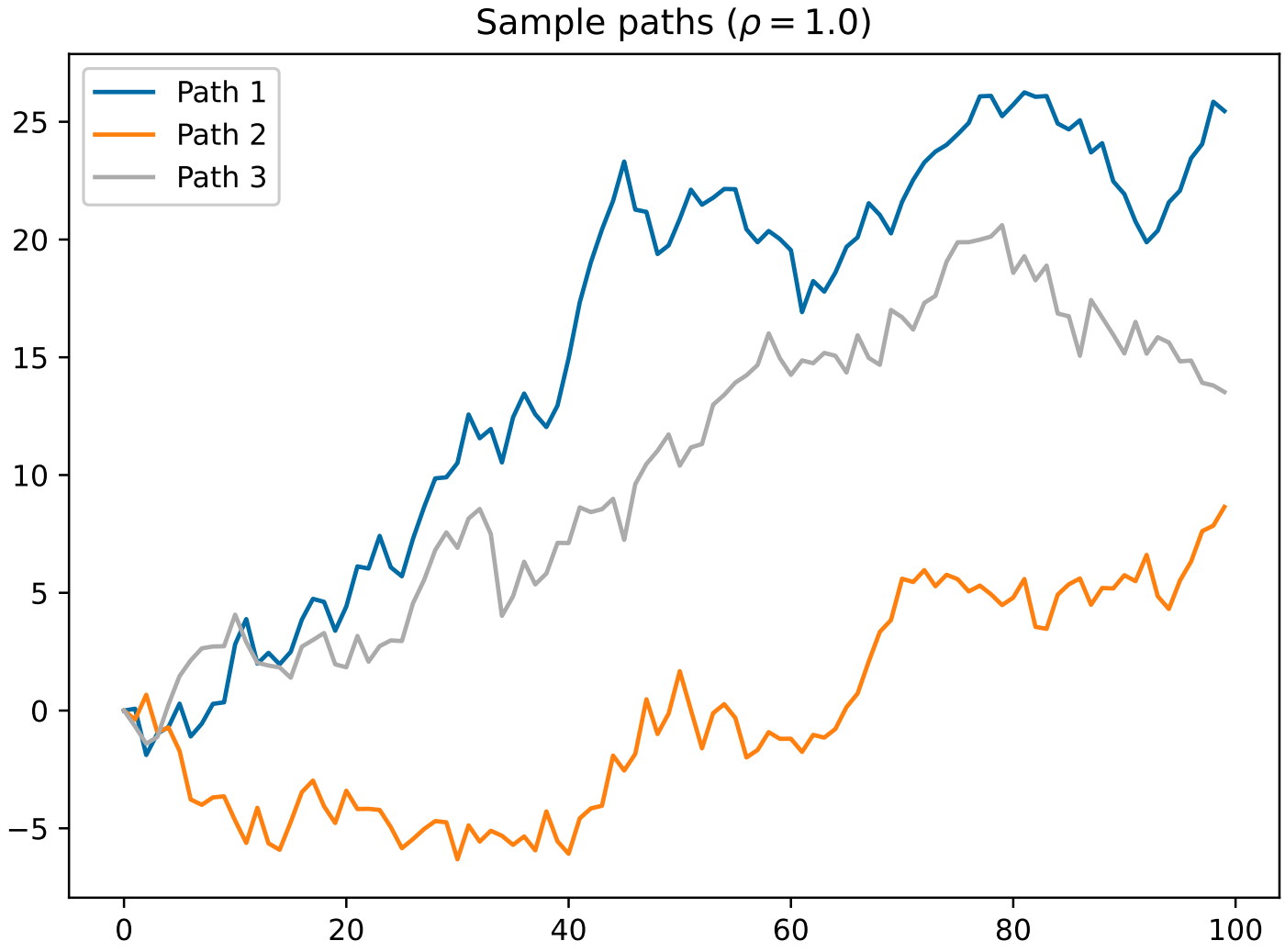
  where $\sigma_u^2 \equiv \text{Var}(u_t)$.

- Variances must be positive. So covariance stationarity requires $\boxed{-1 < \rho < 1}$.



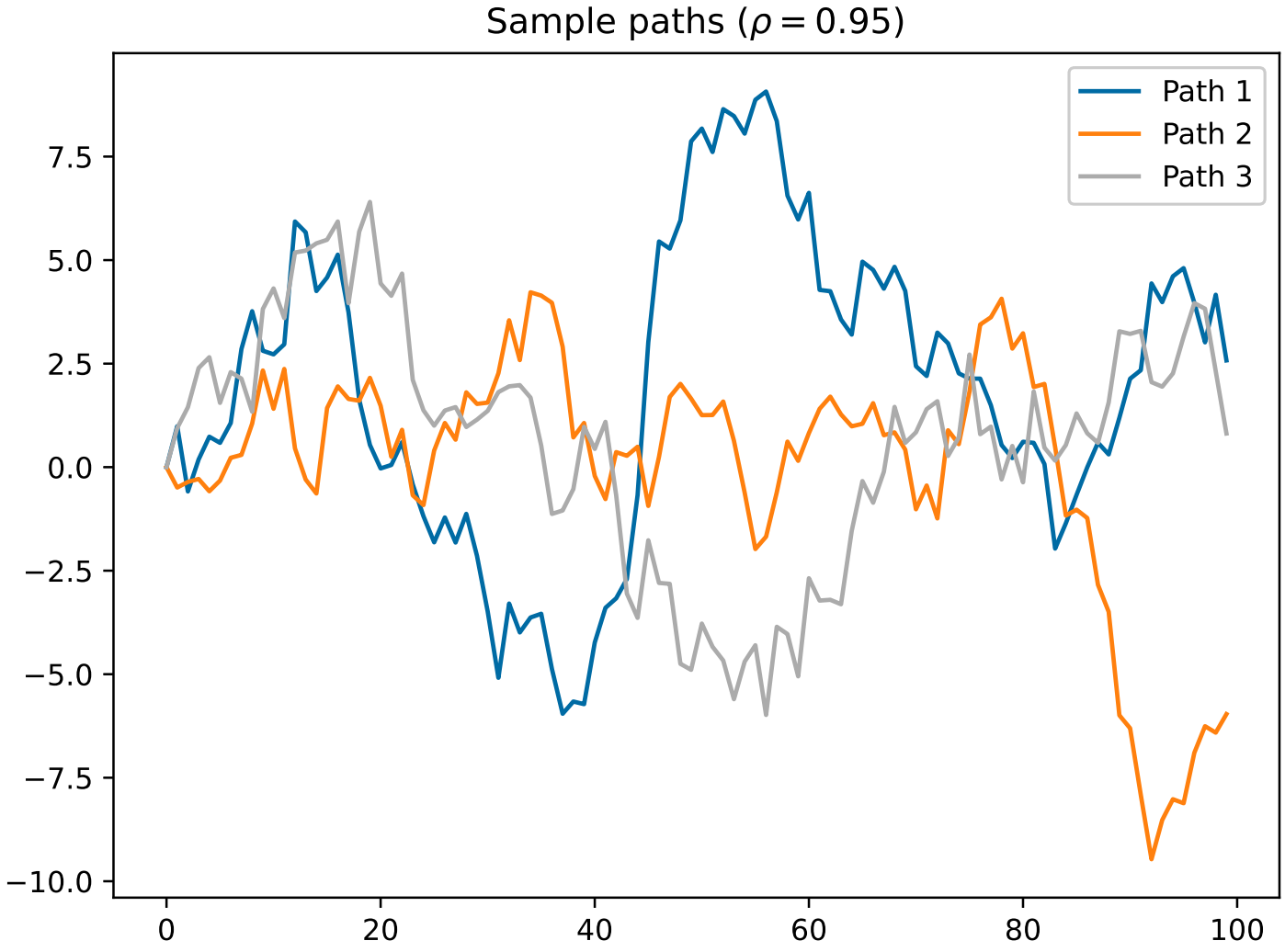
$$y_t = \rho y_{t-1} + u_t,\ u_t \overset{iid}{\sim} N(0,1),\ \rho = 1.1$$

Explosive behavior $\Rightarrow$ Var(y_t) higher at end of sample $\Rightarrow$ non-stationary



$$y_t = \rho y_{t-1} + u_t,\ u_t \overset{iid}{\sim} N(0,1),\ \rho = 1$$

Random walk $\Rightarrow$ doesn't mean-revert $\Rightarrow$ non-stationary



$$y_t = \rho y_{t-1} + u_t,\ u_t \overset{iid}{\sim} N(0,1),\ |\rho| < 1$$

Persistent cycles, but ultimately reverts to mean

AR(1): mean and ACF

- We derived

$$\text{Var}(y_t) = \frac{\sigma_u^2}{1-\rho^2}$$

for a covariance stationary AR(1) process with $-1 < \rho < 1$.

- Similar calculations show (exercise: verify)

$$\boxed{E(y_t) = \frac{c}{1-\rho}}, \quad \boxed{\rho(\ell) \equiv \text{Corr}(y_t, y_{t-\ell}) = \rho^{\ell}},$$

under stationarity.

AR(1): initial condition

- For the AR(1) process to be covariance stationary, we also need the **initial condition** y_0 to have these moments:

$$E(y_0) = \frac{c}{1-\rho}, \quad \text{Var}(y_0) = \frac{\sigma_u^2}{1-\rho^2}.$$

- Then the previous calculations imply that y_t is covariance stationary.
  - If $u_t \overset{iid}{\sim} N(0, \sigma_u^2)$ and $y_0 \sim N\left(\frac{c}{1-\rho}, \frac{\sigma_u^2}{1-\rho^2}\right)$ (independently of the u_t's), then the AR(1) process $\{y_t\}_{t=0}^{\infty}$ is also strictly stationary.
- If y_0 has some other moments, then $\{y_t\}$ will not be covariance stationary, but still,

$$\lim_{t\to\infty} E(y_t) = \frac{c}{1-\rho}, \quad \lim_{t\to\infty} \text{Var}(y_t) = \frac{\sigma_u^2}{1-\rho^2},$$

provided $-1 < \rho < 1$.

AR(1): forecasting

$$y_t = c + \rho y_{t-1} + u_t$$

- Best guess of next-period observation, given past:

$$E[y_{t+1} \mid y_t, y_{t-1}, \ldots, y_1] = E[y_{t+1} \mid y_t] = c + \rho y_t + \underbrace{E[u_{t+1} \mid y_t]}_{=0}.$$

- Best **one-step-ahead** forecast: $\boxed{y_{t+1|t} \equiv c + \rho y_t}$.
- In practice, we need to estimate c and ρ to form a feasible forecast (more soon).

AR(1): multi-step forecast

- By iterating on the AR(1) equation:

$$\begin{aligned} y_t &= c + \rho[c + \rho y_{t-2} + u_{t-1}] + u_t \\ &= c + \rho[c + \rho(c + \rho y_{t-3} + u_{t-2}) + u_{t-1}] + u_t \\ &= \dots \\ &= \rho^h y_{t-h} + \sum_{\ell=0}^{h-1} \rho^\ell (c + u_{t-\ell}). \end{aligned}$$

- Best h-**step-ahead** forecast:

$$\boxed{y_{t+h|t} \equiv E[y_{t+h} \mid y_t] = \rho^h y_t + \frac{1-\rho^h}{1-\rho} c.}$$

AR(p) model

- Autoregression with p lags, **AR(p)**:

  $$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t, \quad E[u_t \mid \{y_s\}_{s<t}] = 0.$$

  $p + 2$ parameters: c, $a_1, a_2, \ldots, a_p$, and σ_u^2.

- If covariance stationary (verify):

$$\boxed{E(y_t) = \frac{c}{1 - (a_1 + a_2 + \cdots + a_p)}}.$$

- What is the autocovariance function $\gamma(\ell) \equiv \text{Cov}(y_t, y_{t-\ell})$?

- (Note: We will postpone until a later slide deck the question of when $\{y_t\}$ is actually covariance stationary.)

AR(p): ACF

$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t, \quad E[u_t \mid \{y_s\}_{s<t}] = 0$$

- Assuming covariance stationarity, if we take covariance with respect to $y_{t-\ell}$ on both sides of the above equation, we get

$$\boxed{\gamma(\ell) = \sum_{j=1}^{p} a_j \gamma(\ell - j) + \mathbb{1}(\ell = 0)\sigma_u^2}. \qquad (\star)$$

- If we stack these equations for $\ell = 0, 1, \ldots, p$, we get a system of $p+1$ linear equations (the **Yule-Walker equations**) in $p+1$ unknowns: $\gamma(0), \gamma(1), \ldots, \gamma(p)$.
  - Allows us to solve for the unknowns given the parameters $a_1, \ldots, a_p, \sigma_u^2$.
- We can then compute $\gamma(\ell)$ for $\ell > p$ by iterating on $(\star)$.
  - We will soon learn a simpler computational method.

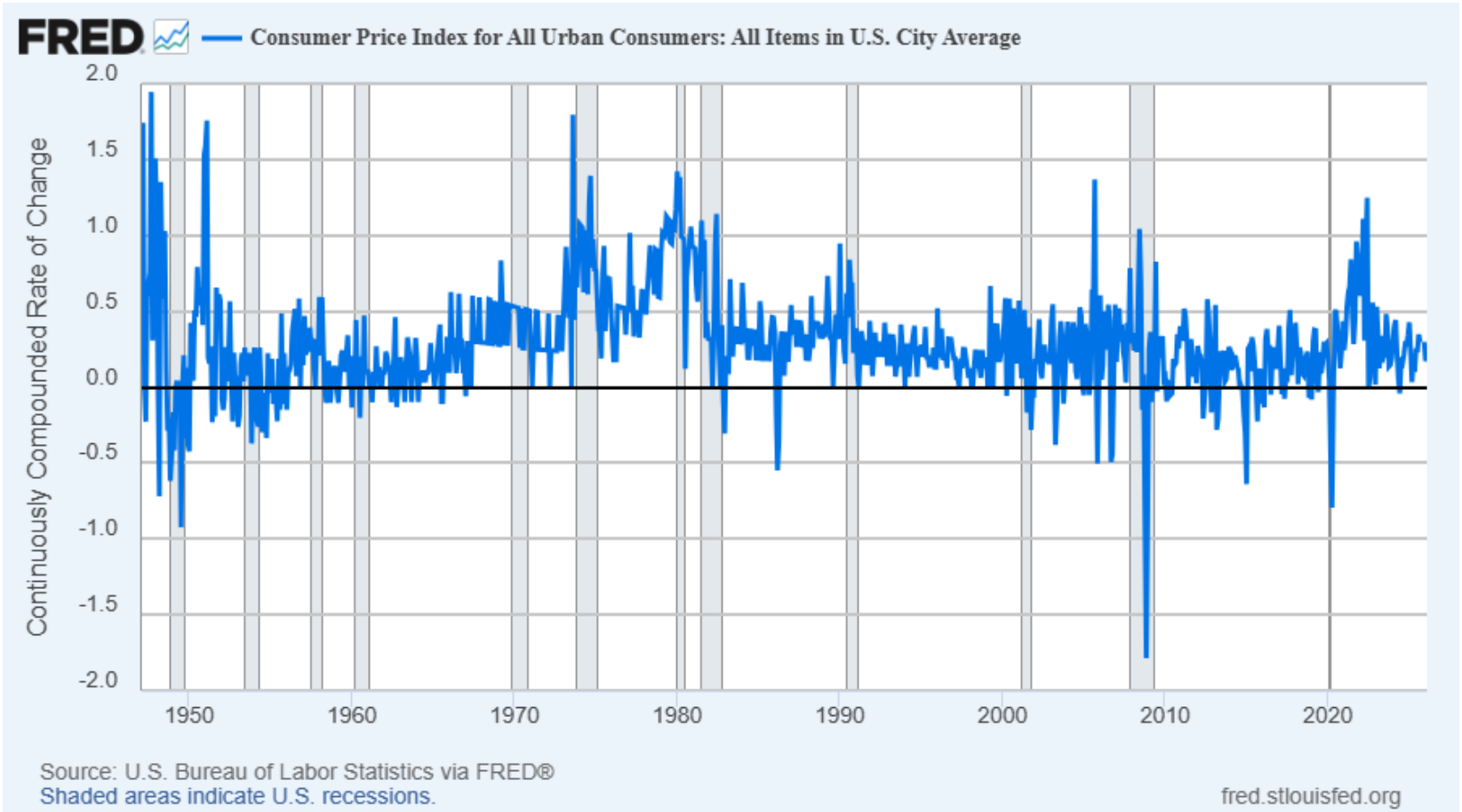
Estimation

$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t, \quad E[u_t \mid \{y_s\}_{s<t}] = 0$$

- Given data $y_1, \ldots, y_T$, we can estimate $c, a_1, \ldots, a_p$ by linear regression (OLS):

$$y_t = \hat{c} + \hat{a}_1 y_{t-1} + \hat{a}_2 y_{t-2} + \cdots + \hat{a}_p y_{t-p} + \hat{u}_t.$$

- The p lags are simply p covariates, as in any other linear regression.
- $\hat{\sigma}_u^2 \equiv \frac{1}{(T-p)-(p+1)} \sum_{t=p+1}^{T} \hat{u}_t^2$.
- Due to the lags, the regression sample starts at $t = p + 1$. The effective sample size for the regression is $T - p$, not T.


FRED
Consumer Price Index for All Urban Consumers: All Items in U.S. City Average
Continuously Compounded Rate of Change
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
fred.stlouisfed.org

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

# Load data
df = pd.read_csv('data/CPIAUCSL.csv', index_col='observation_date',
    parse_dates=True)
df = df.interpolate() # interpolate missing value in Oct 2025
cpi = df['CPIAUCSL'].values
y = 100*np.diff(np.log(cpi)) # monthly inflation rate

# AR order
p = 3

# Build regressor matrix: columns are lags of y
X = np.column_stack([y[p-j:-j] for j in range(1, p + 1)])
X = sm.add_constant(X)
Y = y[p:]

# OLS estimation
res = sm.OLS(Y, X).fit(cov_type='HC1')
xnames = ['const'] + [f'lag{j}' for j in range(1, p + 1)]
print(res.summary(xname=xnames, slim=True))
print(f'\nResidual std dev: {np.sqrt(res.mse_resid):.4f}')
```

OLS Regression Results

Dep. Variable:	y	R-squared:	0.367
Model:	OLS	Adj. R-squared:	0.365
No. Observations:	946	F-statistic:	69.29
Covariance Type:	HC1	Prob (F-statistic):	1.71e-40

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0888	0.015	5.933	0.000	0.059	0.118
lag1	0.5050	0.055	9.192	0.000	0.397	0.613
lag2	0.0250	0.053	0.476	0.634	-0.078	0.128
lag3	0.1584	0.046	3.428	0.001	0.068	0.249

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Residual std dev: 0.2677

OLS estimator: econometric properties

- Is the OLS estimator a good estimator?
  1. Is it consistent?
  2. Is it unbiased?
  3. What is its variance and its distribution?

Consistency

- Question #1: is the estimator **consistent** as $T \to \infty$?
- Focus on AR(1) model for now, without intercept (everything generalizes): $y_t = \rho y_{t-1} + u_t$.
- OLS estimator satisfies

$$\hat{\rho} \equiv \frac{\sum_{t=2}^{T} y_t y_{t-1}}{\sum_{t=2}^{T} y_{t-1}^2} = \frac{\sum_{t=2}^{T} (\rho y_{t-1} + u_t) y_{t-1}}{\sum_{t=2}^{T} y_{t-1}^2}$$
$$= \rho + \frac{\sum_{t=2}^{T} u_t y_{t-1}}{\sum_{t=2}^{T} y_{t-1}^2} = \rho + \frac{\frac{1}{T}\sum_{t=2}^{T} u_t y_{t-1}}{\frac{1}{T}\sum_{t=2}^{T} y_{t-1}^2}.$$

- In your first econometrics course with i.i.d. data, you proved consistency by applying the Law of Large Numbers (LLN) to the numerator and denominator.

Consistency (cont.)

$$\hat{\rho} = \rho + \frac{\frac{1}{T}\sum_{t=2}^{T} u_t y_{t-1}}{\frac{1}{T}\sum_{t=2}^{T} y_{t-1}^2}$$

- In the AR(1) model, the data $\{y_t\}$ is not independent over time. So the usual LLN does not apply.
- As long as the data is stationary ($|\rho| < 1$), it turns out that the LLN applies unchanged (proof: pset question):

$$\frac{1}{T}\sum_{t=2}^{T} u_t y_{t-1} \xrightarrow{p} E[u_t y_{t-1}] = 0 \quad \text{(why?)},$$

$$\frac{1}{T}\sum_{t=2}^{T} y_{t-1}^2 \xrightarrow{p} E(y_{t-1}^2).$$

- By the cts. mapping theorem, the estimator is consistent if $|\rho| < 1$:

$$\hat{\rho} \xrightarrow{p} \rho \quad \text{as} \quad T \to \infty.$$

Bias

- Question #2: is the OLS estimator **unbiased** in finite samples?
  - We know from your first econometrics course that it is for i.i.d. data, if the error term is mean-independent of the regressors.
  - But that proof does not apply here!

$$
\begin{aligned}
E[\hat{\rho}] &= E[E(\hat{\rho} \mid y_1, y_2, \ldots, y_{T-1})] \\
&= \rho + E\left[\frac{\frac{1}{T}\sum_{t=2}^{T} E[u_t \mid y_1, \ldots, y_{T-1}] y_{t-1}}{\frac{1}{T}\sum_{t=2}^{T} y_{t-1}^2}\right]. \qquad (\S)
\end{aligned}
$$

- The error term is **predetermined** by assumption:

$$E[u_t \mid y_{t-1}, y_{t-2}, \ldots, y_1] = 0.$$

- But it's not **strictly exogenous**:

$$E[u_t \mid y_1, \ldots, y_{T-1}] = u_t \neq 0 \quad \text{for} \quad t \leq T-1.$$

So the expectation of the fraction in $(\S)$ is generally non-zero.

Bias (cont.)

- In fact, the OLS estimator (with an intercept) for the AR(1) model displays a *downward* bias in small samples:

$$E[\hat{\rho}] - \rho \approx -\frac{1+3\rho}{T}.$$

- The bias vanishes as $T \to \infty$ (after all, $\hat{\rho}$ is consistent).
- But it can be substantial in small samples, especially when $\rho \approx 1$.
- In AR(p) models, OLS is also biased in small samples and tends to indicate that the data is less persistent than it really is.

Asymptotic normality

- Question #3: what's the distribution of the OLS estimator?
- If the data were i.i.d., the Central Limit Theorem (CLT) would tell us

$$\sqrt{T}\frac{1}{T}\sum_{t=2}^{T} u_t y_{t-1} \xrightarrow{d} N\left(0, E[u_t^2 y_{t-1}^2]\right). \qquad (\sharp)$$

- But our data is not i.i.d. However, the product $u_t y_{t-1}$ is a MDS:

$$E[u_t y_{t-1} \mid y_{t-1}, y_{t-2}, \ldots] = y_{t-1} \times E[u_t \mid y_{t-1}, y_{t-2}, \ldots] = 0.$$

- It turns out that the CLT can be generalized to allow for MDS's, so $(\sharp)$ still applies.

Standard errors and testing

- With the CLT in hand, Slutsky's Lemma implies

$$\sqrt{T}(\hat{\rho} - \rho) = \frac{\sqrt{T}\frac{1}{T}\sum_{t=2}^{T} u_t y_{t-1}}{\frac{1}{T}\sum_{t=2}^{T} y_{t-1}^2} \xrightarrow{d} \frac{N\left(0, E[u_t^2 y_{t-1}^2]\right)}{E(y_{t-1}^2)}.$$

- Loosely:

$$\boxed{\hat{\rho} \overset{approx}{\sim} N\left(\rho, \frac{1}{T} \times \frac{E[u_t^2 y_{t-1}^2]}{\left(E[y_{t-1}^2]\right)^2}\right).}$$

- You may not remember, but this is exactly the same variance expression as in the case of i.i.d. data – though the proof is different!
- Practical upshot: we can use the same formulas for standard errors, t-tests, and F-tests as you learned in your first econometrics course.

OLS Regression Results

Dep. Variable:	y	R-squared:	0.367
Model:	OLS	Adj. R-squared:	0.365
No. Observations:	946	F-statistic:	69.29
Covariance Type:	HC1	Prob (F-statistic):	1.71e-40

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0888	0.015	5.933	0.000	0.059	0.118
lag1	0.5050	0.055	9.192	0.000	0.397	0.613
lag2	0.0250	0.053	0.476	0.634	-0.078	0.128
lag3	0.1584	0.046	3.428	0.001	0.068	0.249

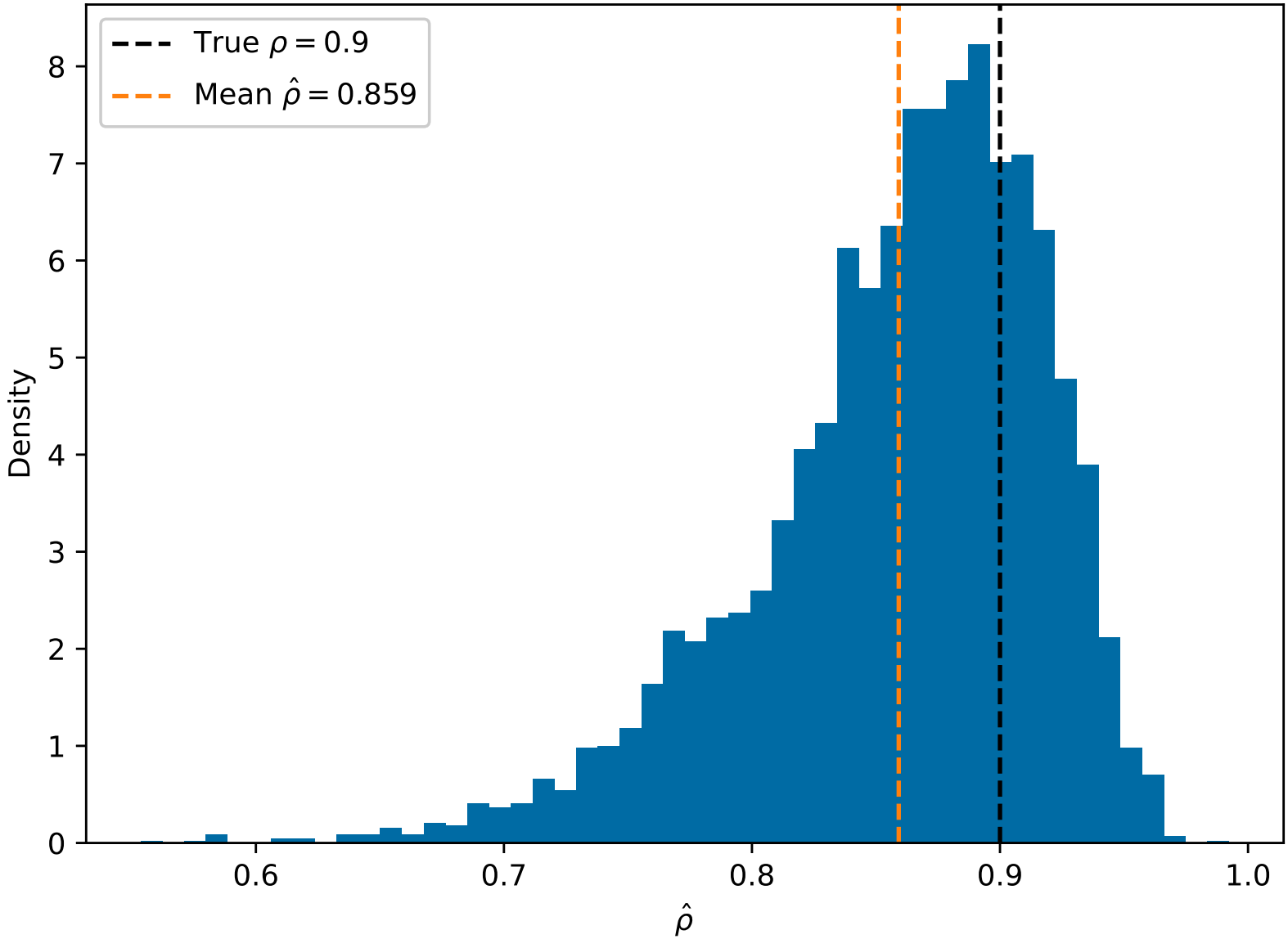
Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

- 95% confidence interval for a_1: $[0.397, 0.613]$
- Coefficient on 2nd lag is insignificant at 10% level.
- F-test rejects that process is white noise w. p-value $< 2 \times 10^{-40}$.

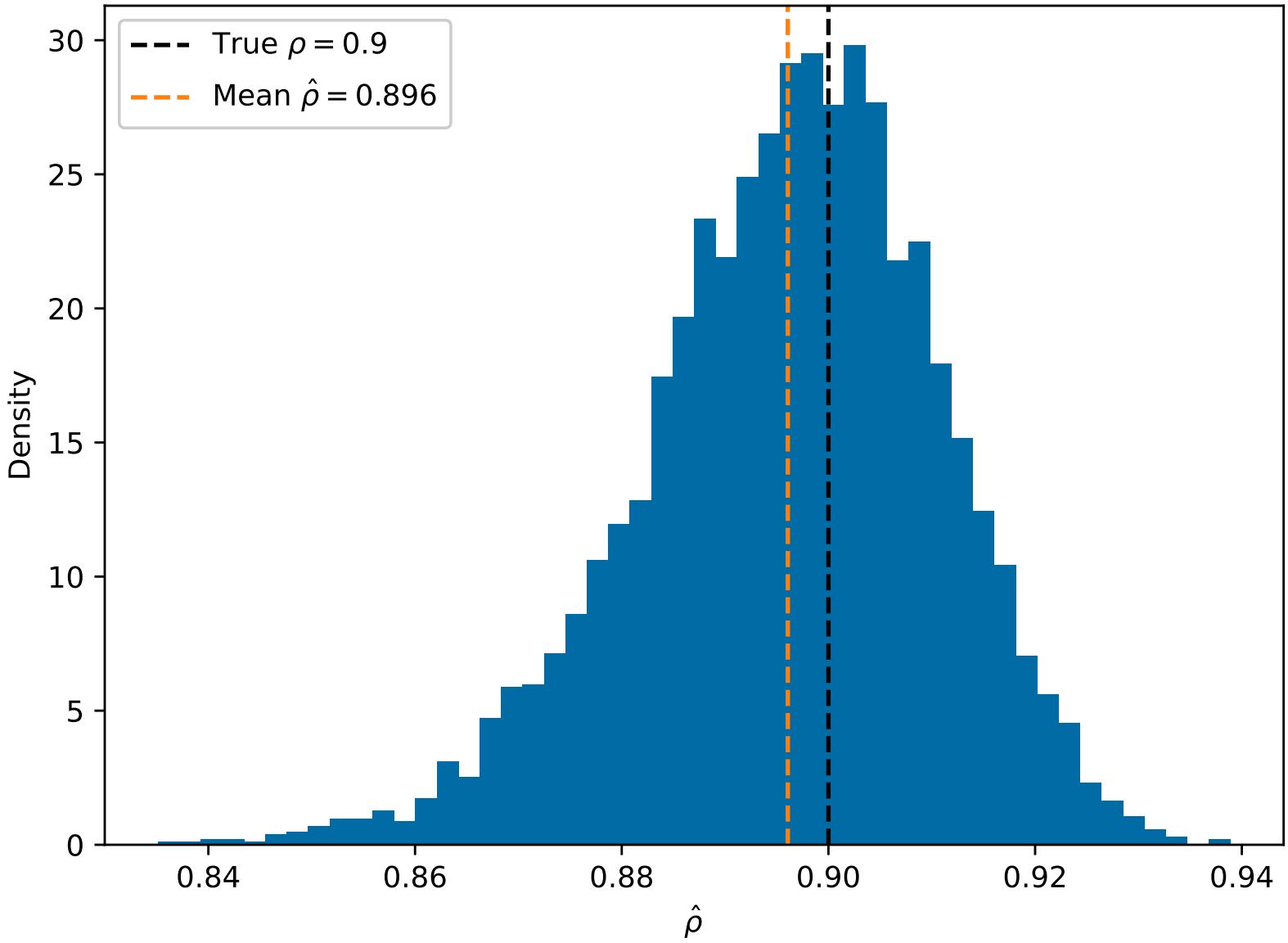
Simulation illustration of econometric properties

```
# Parameters
rho = 0.9
T = 100
n_sim = 5000
rng = np.random.default_rng(seed=31150)

# Monte Carlo loop
rho_hat = np.zeros(n_sim)
for i in range(n_sim):
    # Simulate AR(1)
    u = rng.standard_normal(T)
    y = np.zeros(T)
    for t in range(1, T):
        y[t] = rho * y[t - 1] + u[t]
    # OLS estimation
    X = sm.add_constant(y[:-1])
    Y = y[1:]
    res = sm.OLS(Y, X).fit()
    rho_hat[i] = res.params[1]
```


True $\rho = 0.9$
Mean $\hat{\rho} = 0.859$
Density
0
1
2
3
4
5
6
7
8
0.6
0.7
0.8
0.9
1.0
$\hat{\rho}$

$T = 100$



$T = 1000$ (notice different horizontal axis limits)

Lag order selection

- How many lags p to include in our AR(p) regression?
- Bias-variance trade-off:
  - More lags: better fit complicated dynamics $\Rightarrow$ lower bias.
  - Fewer lags: fewer parameters to estimate $\Rightarrow$ lower variance.

Lag order selection: general-to-specific

- One strategy for picking p: **general-to-specific testing**.
  1. Start with large p.
  2. Test whether coefficient on p-th lag is zero at signif. level α.
  3. If don't reject, estimate AR($p-1$) model.
  4. Test whether the last coefficient is zero.
  5. Continue until we reject, then stop.
- This strategy has probability of at least α (in practice, often much higher) of selecting too large a model, since every test in the sequence may falsely reject (type-I error).

Lag order selection: BIC

- Instead, it's common to select p to minimize the **Bayesian information criterion** (BIC):

$$BIC(p) \equiv \log\{\hat{\sigma}(p)^2\} + p \times \frac{\log T_{\text{eff}}}{T_{\text{eff}}}.$$

  - $\hat{\sigma}(p)^2 \equiv$ sample variance of OLS residuals from AR(p) regression.
  - $T_{\text{eff}} \equiv$ effective sample size (# obs. in the regression).

- This criterion intuitively captures the bias-variance trade-off:
  - Larger p improves fit so decreases first term.
  - Second term penalizes complex models with large p.

- The BIC is **consistent**: if data is generated by an AR(p_0), then

$$P\left(p_0 \in \text{argmin}_p\, BIC(p)\right) \to 1 \quad \text{as} \quad T \to \infty.$$

Lag order selection: AIC

- A variant of the BIC is the **Akaike information criterion** (AIC):

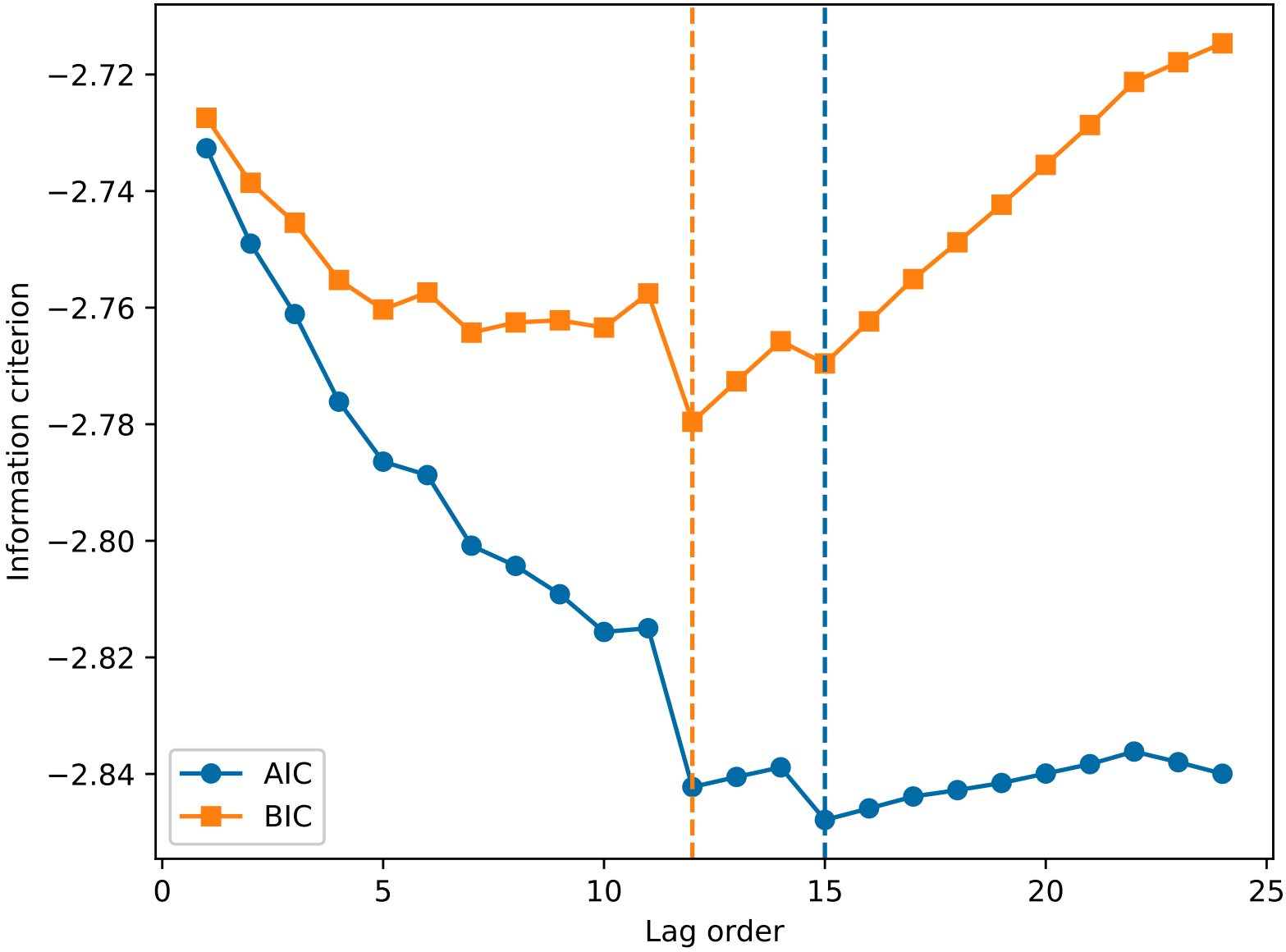
$$AIC(p) \equiv \log\{\hat{\sigma}(p)^2\} + p \times \frac{2}{T_{\text{eff}}}.$$

- When $T_{\text{eff}} \geq 8$ (so $\log(T_{\text{eff}}) > 2$), AIC penalizes complex models less than BIC, so selects (weakly) larger models.
- As a result, AIC is *inconsistent*: positive probability of selecting too large models, even when T is very large.
- However, AR forecasts made with p selected by AIC are **efficient**: when the true model has ∞-ly many lags, the forecast errors tend to be smaller on average than those made with any other lag order selection method, at least when T is large.
- Dilemma: consistency or efficiency? Since AR models are at best approximations, efficiency seems more relevant.

```
p_max = 24 # max lag order
p_grid = np.arange(1, p_max + 1)
Y = y[p_max:]
T_eff = len(Y)
aic = np.zeros(p_max)
bic = np.zeros(p_max)

for i, p in enumerate(p_grid):
    # Build regressor matrix (enforce same sample for all p)
    X = np.column_stack([y[p_max-j:-j] for j in range(1, p + 1)])
    X = sm.add_constant(X)
    # OLS estimation
    res = sm.OLS(Y, X).fit()
    sigma2 = np.var(res.resid)
    # Information criteria
    aic[i] = np.log(sigma2) + 2 * p / T_eff
    bic[i] = np.log(sigma2) + np.log(T_eff) * p / T_eff
```

Note: it's conventional to run all AR regressions on the same sample $t = p_{\max} + 1, p_{\max} + 2, \ldots, T$, so $T_{\text{eff}} = T - p_{\max}$.


−2.72
−2.74
−2.76
−2.78
−2.80
−2.82
−2.84
Information criterion
AIC
BIC
0
5
10
15
20
25
Lag order

Practical coding

- For pedagogical reasons, my Python code computed AR estimates and information criteria "by hand" from OLS output.
- But in practice, we can use `statsmodels`'s built-in functions for AR modeling:
  - Estimation and inference: `statsmodels.tsa.ar_model.AutoReg`
  - BIC/AIC: `statsmodels.tsa.ar_model.ar_select_order`

What did we learn?

- AR(p) model
  - Estimation by OLS
  - Econometric properties: consistency, bias, asymptotic normality
  - Standard errors, testing
- Lag length selection
  - General-to-specific testing
  - BIC, AIC, consistency vs. efficiency

Looking ahead

- What are the sources of forecast errors?
- How do we evaluate the quality of forecasts?
- Forecast intervals: conveying uncertainty about forecasts